

[8/12/17-9]

SARDAR PATEL UNIVERSITY**B.Sc.(CA&IT) (Regular)/ M.SC(IT) Integrated (NC)****Sem-II EXAMINATION****MONDAY, 10th APRIL, 2017****10:00 am to 12:00 noon****SUBJECT: MATHEMATICS-II (PS02FIIT02)**

Maximum Marks: 70

Q-1 Write the correct option in the answer book.

[10]

(1) 3 boys can be selected from group of 9 boys in..... ways.

- (a) 9! (b) 504 (c) 84 (d) $\frac{9!}{4!}$

(2) The set $\{N, +\}$ is

- (a) group (b) ring (c) monoid (d) semi group

(3) The quartile Q_2 is coincides with

- (a) Mean (b) Mode (c) Median (d) Standard deviation

(4) Let $A = \{0, 1\}$, then A closed under:

- (a) multiplication (b) addition (c) Division (d) Subtraction

(5) How many ways the judge can award first and second prizes among 10 contestants?

- (a) 100 (b) 90 (c) 10 (d) 45

(6) The number of elements in a set $\{x \in Z: x^2 - 2 = 0\}$ are:

- (a) 1 (b) 2 (c) $\pm\sqrt{2}$ (d) 0

(7) The set $\{x \in Z: -75 < x < 75\}$ is:

- (a) finite (b) Infinite (c) Empty (d) none

(8) If the regression coefficient $b_{XY} > 1$, then

- (a) $b_{YX} = 0$ (b) $b_{YX} > 0$ (c) $b_{YX} < 1$ (d) None of these

(9) De Morgan's Law:

- (a) $(A \cup B)^c = A^c \cap B^c$ (b) $(A \cap B)^c = (A \cap B)^c$
(c) $(A \cup B)^c = A \cap B$ (d) None

(10) Every monoid are

- (a) group (b) ring (c) semi group (d) none

Q:2 Answer the following in short. (Any Ten)

[20]

(1) Find the number of distinct words that can be formed from all the letters of the word (1) ANAND (2) GUJARAT. (PTO)

- (2) In Z_{10} , find -3 , -8 , 3^{-1} , 5^{-1} .
- (3) Explain the positive correlation with two examples.
- (4) How many committees of six with a given chair person can be selected from ten persons?
- (5) Find the power set of a set $\{a, b, c\}$.
- (6) Define Semi group and Monoid.
- (7) Find variance of the observations: 3, 3, 3, 3, 3, 3, 3, 3.
- (8) Find dual of the following:
(i) $(A \cap B \cup C)^c = (A \cap C)^c \cap (A \cap B)^c$, (ii) $(A \cap \phi) \cup (U \cap A^c) = A$.
- (9) Find the number of ways that a party of 7 persons can arrange themselves: (i) around a circular table and (ii) in a row.
- (10) If S is a nonempty set with the operation $a*b = b$. Is the operation* associative?
- (11) Define range and variance.
- (12) If $A = \{1, 2, 3, \dots, 8\}$ and $B = \{4, 6, 7, \dots, 14\}$ then find the symmetrical difference between A and B .

Q-3

- (a) Let n denote a positive integer. Suppose a function L is defined as [05]

$$L(n) = \begin{cases} 0 & \text{if } n=1 \\ L([n/2]) + 1 & \text{if } n > 1 \end{cases} \text{ Find } L(25) \text{ and } L(34).$$

- (b) Find inverse of the function $f(x) = \frac{7x-3}{5x-2}$, $x \neq \frac{2}{5}$. [05]

Q-3

OR

- (c) Prove that $1+2+3+\dots+n = \frac{n(n+1)}{2}$. [05]

- (d) If $f(x) = x^2 + 7$ and $g(x) = 2x - 5$ then find: (i) $f \circ g$ (ii) $f \circ g(-1)$ (iii) $g \circ f(1)$. [05]

Q-4

- (a) For a, b rational number set Q , define $a*b = ab/3$. Show that $(Q, *)$ is a group under binary operation $*$. Is it commutative? [05]

- For a, b rational number, define $a*b = a + b - ab$. Is $(Q, *)$ commutative? Show that [05]
(b) $(Q, *)$ is Monoid and find its inverse if it exist.

Q-4

OR

- (c) Show that the set $\{1, 2, 3, 4, 5, 6\}$ is a group under multiplication modulo 7. [05]

- (d) For $a, b \in Q$ (rational numbers), define $a*b = a + b + ab$. [05]

- (i) Is $(Q, *)$ Semi group?
- (ii) Is $(Q, *)$ Monoid ?
- (iii) Find it's inverse if it exist.

Q-5

- (a) A debating team consists of 3 boys and 3 girls. Find the number of ways they can sit in a row where: (a) there are no restrictions; (b) the boys and girls are each to sit together; (c) just the girls are to sit together. [05]

- (b) Find the number m of five letter "words" containing two different vowels and three different consonants that can be formed from the 26 alphabets. Also find m if the words must begin with B. [05]

Q-5

OR

- (c) A class consist of seven men and five women. (i) Find the number m of committees of five that can be selected from the class (ii) Find the number m of committees of five if it consists of three men and two women. [05]

- (d) Suppose repetitions are not permitted. (a) Find the number of three-digit numbers that can be formed from the six digits 2, 3, 5, 6, 7, and 9. (b) How many of them are less than 400? (c) How many of them are even? [05]

- Q.6 Calculate the correlation coefficient for the following heights (in inches) of fathers (X) and their sons (Y): [05]

X:	65	66	67	67	68	69	70	72
Y:	67	68	65	68	72	72	69	71

- (b) Explain the linear regression and state its properties. [05]

OR

- Q.6 Following table shows the frequency distribution of marks. [05]

Marks :	0- 10	10-20	20-30	30-40	40-50	50-60
No. of students:	12	18	27	20	17	6

Find the Mean deviation from mean for above frequency distribution.

- (b) Frequency distribution of the blood pressure given below, Compute the quartile Deviation of the frequency distribution. [05]

Systolic BP (mm Hg)	10-20	20-30	30-40	40-50	50-60	60-70	70-80	80-90
No. of Infants	1	6	14	43	21	13	10	1

— X — X —

(3)

